

Industrial Roof Leakage Prevention Checklist

A project-stage checklist for reviewing slope, drainage, flashings, penetrations, gutters, and installation risk before roofing starts.

CHECKLIST | Technical Guides | 8 pages | March 2026

How to Use This Checklist

This checklist is completed before roofing installation begins. It is a pre-mobilization quality tool that helps the buyer, contractor, and project engineer confirm that slope, drainage, flashing, penetration, and installation decisions are resolved on paper before any panel is fixed. Catching a design or coordination gap at this stage costs a review meeting; catching it after handover costs a roof.

Three usage rules

1. Complete all sections before mobilization to site.
2. Flag any failed (X) item to the project engineer before proceeding — do not start installation with open items.
3. File the signed checklist in the project quality record as evidence of pre-start review.

Section 1: Slope & Drainage Design

Item	Pass ✓	Fail ✗	Notes
Roof slope is minimum 3° for sandwich panels, 1° for standing seam.			
Drainage direction confirmed on shop drawings.			
No low points or valleys without a drainage outlet.			
Internal gutters sized for peak rainfall intensity.			
External gutters have sufficient fall to downpipes.			
Downpipe positions coordinated with structural columns.			
Anti-ponding strips specified at eaves.			
Roof plan reviewed for water accumulation risk zones.			
Expansion joints include drainage provision.			
Drainage design reviewed by the structural engineer.			

Section 2: Flashing Design

Item	Pass ✓	Fail ✗	Notes
Ridge flashing detail confirmed on shop drawings.			
Eave flashing with drip edge specified.			
Wall-to-roof junction flashing detailed.			
Valley flashing (if applicable) specified with sealant.			
Parapet flashing with cap and cleat shown.			
Flashing material compatible with panel skin material.			
Flashing overlap minimum 150 mm confirmed.			
Sealant type specified (neutral-cure silicone or equivalent).			
All flashing fixings are concealed or sealed.			
Thermal movement allowance included in flashing design.			

Section 3: Penetrations

Item	Pass ✓	Fail ✗	Notes
All roof penetrations identified on the plan.			
Each penetration has a detail drawing.			
Penetration collars or curbs specified.			
Minimum upstand height 150 mm above roof level.			
Sealant schedule matches penetration material type.			
HVAC penetrations have vibration isolation.			
Electrical conduit penetrations are waterproofed below deck.			
Penetration locations avoid panel ridge lines.			

Section 4: Installation Risk Review

Item	Pass ✓	Fail ✗	Notes
Installation sequence matches shop-drawing panel numbering.			
Panel storage on site is off the ground and covered.			
Panels are not stored in direct sunlight for extended periods.			
Self-drilling fastener torque specification confirmed.			
Fastener spacing matches engineering design.			
No panel cutting over installed panels (debris contamination).			
Protective film removed within 30 days of installation.			
Sealant application temperature within manufacturer range.			
No foot traffic on installed panels without walk pads.			
Post-installation inspection checklist planned.			

Section 5: Gutters & Downpipes

Item	Pass ✓	Fail ✕	Notes
Gutter profile and size shown on drawings.			
Gutter fall minimum 1:500 confirmed.			
Expansion joints in gutters at maximum 15 m spacing.			
Gutter brackets at maximum 900 mm centers.			
Downpipe diameter sufficient for roof catchment area.			
Downpipe discharge location confirmed (drain, soakaway, or open).			
Bird / leaf guards specified for open gutters.			

Sign-off

Final engineering responsibility remains with the project engineer of record. This checklist is provided as a technical aid.

Project Name	
Project Location	
Date	
Reviewed by	
Role	
Signature	
SIPANEL Reference	